Parul Singh

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PROFILE SUMMARY

AI/ML Engineer with experience in building and deploying intelligent applications using machine learning, NLP, and deep learning. Skilled in developing end-to-end AI solutions, including chatbots, recommendation systems, and computer vision models.

EDUCATION

BTech (CSE)- Chhatrapati Shahu Ji Maharaj University, Kanpur (CPI: 8.62/10) 2021-2025

EXPERIENCE

Indian Institute of Technology, Kanpur IITK Campus, Kanpur

Research Intern, May 2024 - July 2024

- Developed a Django-based automation system integrating multiple regulatory sources, eliminating manual document search and saving ~20 hours/week
- Built a scalable pipeline to fetch, process, and store regulatory documents using web scraping techniques
- Designed backend workflows for automated updates, improving efficiency and reducing compliance risks
- Collaborated in a team to deliver a production-ready solution for real-world use

PROJECTS

AI Career Coach **GitHub Link**

- Built a RAG-based AI application for personalized career guidance, resume evaluation, and skill gap analysis
- Implemented Retrieval-Augmented Generation using LangChain, FAISS, and LLM APIs for context-aware responses
- Designed Flask-based backend to handle dynamic queries and document processing
- Deployed the application on Render for real-time access and scalability
- **Tech Stack:** Python, Flask, LangChain, FAISS, LLM APIs (Llama-3/Groq), Embeddings, Tailwind CSS, Render

QueryPDF – AI-Powered Document Assistant **GitHub Link**

- Built an AI-powered system to query PDF documents using natural language with a RAG-based pipeline
- Implemented semantic search using FAISS and embeddings for accurate information retrieval
- Integrated Groq LLM (LLaMA models) with LangChain for context-aware responses
- Built an end-to-end pipeline from document scraping to AI-based querying
- **Tech Stack:** Python, Flask, LangChain, FAISS, Groq API (LLaMA), HuggingFace Embeddings, Tailwind CSS

Solar Panel Defect Classification System **GitHub Link**

- Built a deep learning model using EfficientNet for multi-class solar defect classification
- Developed a Streamlit interface for image upload and real-time predictions
- Implemented preprocessing and achieved accurate classification across 6 defect categories.
- **Tech Stack:** Python, TensorFlow/Keras, Streamlit, NumPy, Pandas, Plotly

Book Recommender System **GitHub Link**

- Built a recommendation system using collaborative filtering and popularity-based techniques
- Generated personalized recommendations by analyzing user behavior and large datasets
- **Tech Stack:** Python, Pandas, NumPy, Scikit-learn, Streamlit

TECHNICAL SKILLS

- **Languages:** Python, SQL, JavaScript
- **Machine Learning:** Regression, Classification, Feature Engineering, Model Evaluation
- **Deep Learning:** CNN, TensorFlow, Keras
- **AI & NLP:** NLP, Chatbot Development, Prompt Engineering, RAG, LLM Integration
- **Libraries:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, NLTK
- **Web Development**: Django, Flask, HTML, CSS
- **Deployment & Tools:** Streamlit, Render (Model Deployment), Git, GitHub, Jupyter Notebook, Google Colab
- **Data Analysis:** EDA, Data Cleaning, Data Visualization

ACHIEVEMENTS

- Selected for **SURGE Program 2024 (IIT Kanpur)** under institute-funded category
- Selected in Round 1 of **Unstop Playstorm** for presenting *The Glitch Room*